

Microplastic Polymer Analysis Report

Report ID	dc18df42-71e8-4067-8f9b-bee6ab862547
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Status	Completed
Sample ID	HDPEFTIR001

Prediction Results

Detected Polymer	HDPE
Confidence Score	100.00%
Model Used	Peak Identifier

FTIR Spectral Analysis

- 2915 cm^{-1} → CH₂ asymmetric stretching
- 1470 cm^{-1} → CH₂ bending

Confidence Interpretation

■■■■■■■■■■ 100.00%

Scientific References

- [1] Harper, C. Handbook of Plastics Technologies.
- [2] Thompson, R. Plastic debris in oceans.
- [3] Barnes, D. Accumulation of plastics.
- [4] Andrady, A. Environmental impacts of plastics.
- [5] Jung, M. FTIR analysis of HDPE.

AI Generated Summary

The analyzed sample was identified as High-Density Polyethylene (HDPE) with a confidence score of 100.00%. The FTIR spectral characteristics strongly support the presence of HDPE polymer chains. The detected CH2 asymmetric stretching at 2915 cm^{-1} and CH2 bending at 1470 cm^{-1} are characteristic of the hydrocarbon functional groups in HDPE. This polymer is widely used in plastic bags, bottles, and containers. The analysis indicates a reliable classification suitable for environmental assessment and waste management studies.